

Multiple Choice (10 marks)

1. The four big moons of Jupiter are Callisto Europa Ganymede and Io. Which one of them has the smallest distance to Jupiter?
 - (a) Callisto
 - (b) Europa
 - (c) Ganymede
 - (d) Io
2. The lunar highlands are:
 - (a) ancient heavily cratered highlands
 - (b) dark lavas inside volcanic calderas
 - (c) dark lavas filling older impact basins
 - (d) the bright regions on the Moon
3. Which of the following was not cited as evidence for life in the martian meteorite ALH84001?
 - (a) Complex organic molecules specifically PAHs
 - (b) Magnetite grains similar to those formed by bacteria on Earth
 - (c) Carbonate minerals indicating a thicker warmer Martian atmosphere
 - (d) Amino acids with a preferred orientation or "chirality"
4. What about asteroids makes them appear faint in sky surveys searching for them?
 - (a) Asteroids emit pulsed radiation
 - (b) Asteroids emit a lot of their own radiation
 - (c) They reflect enough of the sun's light to make them brighter than most background stars
 - (d) Asteroids have substantial motion relative to the background stars
5. What effect or effects would be least significant if the Moon's orbital plane were exactly the same as the ecliptic plane?
 - (a) Solar eclipses would be much rarer.
 - (b) Solar eclipses would last much longer.
 - (c) Solar eclipses would be much more frequent.
 - (d) Solar eclipses would not last as long.

True / False (10 marks)

Answer True or False

6. True or False: In 20000 years, the Moon will be closer to the Earth and the length of the Earth's day will increase due to tidal forces.
7. True or False: Cassiopeia is a prominent M-shaped constellation located in the northern sky, recognizable by its distinctive shape.
8. True or False: The Great Red Spot is a persistent **cyclonic** storm located in the atmosphere of Jupiter.
9. True or False: Earth is the first planet in the solar system known to have active plate tectonics.
10. True or False: Fresh impact craters on solid surfaces sometimes include central peaks as a defining characteristic.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the concept of a white dwarf in astronomy, explaining its definition and significance in understanding the radiation of stars and their thermal properties.
12. Discuss the relationship between Mars's atmospheric pressure and its magnetic field strength, analyzing how these factors influence the planet's geophysical characteristics.

End of Paper